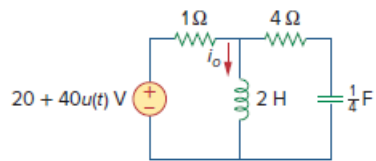


Problem

Determine $i_o(t)$ in the network shown in Fig. 16.70.

Figure 16.70:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 16.70 in the textbook.

It is known that, the step function $u(t)$ is,

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

For $t < 0$, the value of voltage source is 20 V, inductor acts as short circuit and capacitor acts as open circuit.

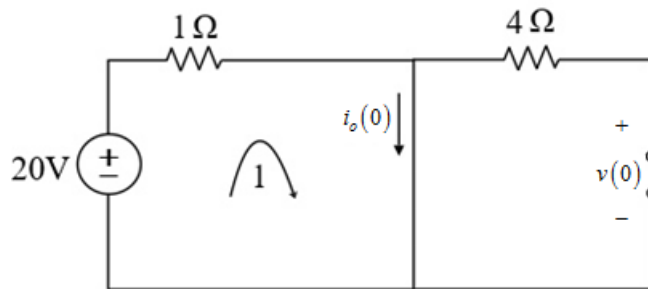


Figure 1

Step 2 of 7

Calculate the initial current through the inductor.

$$\begin{aligned} i_o(0) &= \frac{20 \text{ V}}{1 \Omega} \\ &= 20 \text{ A} \end{aligned}$$

Calculate the initial voltage across the capacitor.

$$v(0) = 0 \text{ V}$$

Calculate the voltage source value for $t > 0$.

$$\begin{aligned} V &= 20 + 40u(t) \\ &= 20 + 40(1) \quad (\text{since } u(t) = 1 \text{ for } t > 0) \\ &= 60 \text{ V} \end{aligned}$$

Step 3 of 7

The frequency domain equivalent circuit for $t > 0$ is as shown in Figure 2.

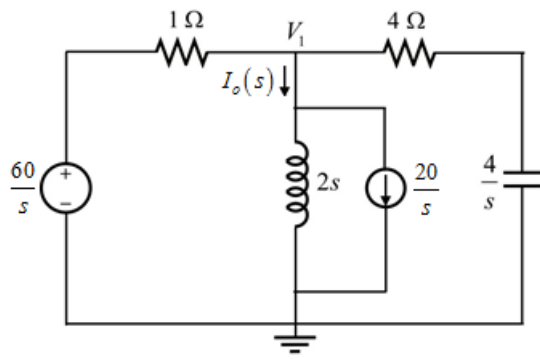


Figure 2

Step 4 of 7

Apply Kirchhoff's current law at node V_1 .

$$\begin{aligned} \frac{V_1 - \frac{60}{s}}{1} + \frac{V_1}{2s} + \frac{20}{s} + \frac{V_1}{4 + \frac{4}{s}} &= 0 \\ \left(1 + \frac{1}{2s} + \frac{s}{4s+4}\right)V_1 &= \frac{60}{s} - \frac{20}{s} \\ \left(\frac{2s(4s+4) + 4s + 4 + 2s^2}{2s(4s+4)}\right)V_1 &= \frac{40}{s} \\ \left(\frac{10s^2 + 12s + 4}{2s(4s+4)}\right)V_1 &= \frac{40}{s} \\ V_1 &= \frac{40(4s+4)}{5s^2 + 6s + 2} \dots\dots (1) \end{aligned}$$

Step 5 of 7

Calculate the current $I_o(s)$.

$$I_o(s) = \frac{V_1}{2s} + \frac{20}{s}$$

Substitute $\frac{40(4s+4)}{5s^2 + 6s + 2}$ for V_1 .

$$\begin{aligned} I_o(s) &= \frac{\left(\frac{40(4s+4)}{5s^2 + 6s + 2}\right)}{2s} + \frac{20}{s} \\ &= \frac{20(4s+4) + 20(5s^2 + 6s + 2)}{s(5s^2 + 6s + 2)} \\ &= \frac{20(5s^2 + 10s + 6)}{s(5s^2 + 6s + 2)} \end{aligned}$$

Step 6 of 7

Apply partial transform.

$$I_o(s) = \frac{A}{s} + \frac{Bs+C}{5s^2+6s+2} \dots\dots (2)$$

$$\frac{20(5s^2+10s+6)}{s(5s^2+6s+2)} = \frac{A(5s^2+6s+2)+Bs^2+Cs}{s(5s^2+6s+2)}$$

$$100s^2+200s+120 = A(5s^2+6s+2)+Bs^2+Cs \dots\dots (3)$$

Equate constant terms from equation (3).

$$120 = 2A$$

$$A = 60$$

Equate s terms from equation (3).

$$200 = 6A + C$$

$$200 = (6)(60) + C$$

$$C = -160$$

Step 7 of 7

Equate s^2 terms from equation (3).

$$100 = 5A + B$$

$$100 = (5)(60) + B$$

$$B = -200$$

Recall equation (2).

$$I_o(s) = \frac{A}{s} + \frac{Bs+C}{5s^2+6s+2}$$

Substitute 60 for s , -200 for B and -160 for C .

$$\begin{aligned} I_o(s) &= \frac{60}{s} + \frac{-200s-160}{5s^2+6s+2} \\ &= \frac{60}{s} - \frac{200s+160}{5(s^2+1.2s+0.4)} \\ &= \frac{60}{s} - \frac{40s+32}{s^2+1.2s+0.36+0.04} \\ &= \frac{60}{s} - \frac{40(s+0.6)}{(s+0.6)^2+(0.2)^2} - \frac{40(0.2)}{(s+0.6)^2+(0.2)^2} \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} i_o(t) &= 60u(t) - 40e^{-0.6t} \cos(0.2t)u(t) - 40e^{-0.6t} \sin(0.2t)u(t) \text{ A} \\ &= [60 - 40e^{-0.6t} \{\cos(0.2t) + \sin(0.2t)\}]u(t) \text{ A} \end{aligned}$$

Therefore, the current $i_o(t)$ is $\boxed{[60 - 40e^{-0.6t} \{\cos(0.2t) + \sin(0.2t)\}]u(t) \text{ A}}.$